Today

Lagrange Multipliers

KKT conditions

Augmented Lagrange

Announcements

Exam grades tomorrow

HW 7 due Thursday

Lagrange Multipliers

$$\mathcal{L}(\vec{x}, \vec{\lambda}, \vec{\sigma}) = f(\vec{x}) + \vec{\lambda} \cdot \vec{h}(\vec{x}) + \vec{\sigma} \cdot \vec{g}(\vec{x})$$

Lagrangian

Equality Lagrange Multipliers

Inequality L.M.

At the constrained optimum

$$\nabla_{\vec{x}} \mathcal{L}(\vec{x}^*, \vec{\lambda}^*, \vec{\sigma}^*) = \vec{0}$$

$\vec{\lambda}^*, \vec{\sigma}^*$ need to be the "right" values — the ones that make $\vec{x}^*$ satisfy the constraints

For the demo example

$$\mathcal{L}(x, \sigma) = x^2 + \sigma(1-x)$$

F.O.N.C.

Unconstrained

$$\nabla_{\vec{x}} f(\vec{x}) = \vec{0}$$

Constrained

$$\nabla_{\vec{x}} \mathcal{L}(\vec{x}^*, \vec{\lambda}^*, \vec{\sigma}^*) = \vec{0} \quad \text{stationarity}$$

$$\vec{h}(\vec{x}^*) = \vec{0}$$

$$\vec{g}(\vec{x}^*) \leq \vec{0}$$

$$\vec{\sigma}^* \geq \vec{0}$$

$$\vec{\sigma}^* \odot \vec{g}(\vec{x}^*) = \vec{0}$$

Karush-Kuhn-Tucker KKT conditions

solving constrained optimization problems is mostly about solving KKT

$\odot$ elementwise multiplication .* in matlab

"Complementary slackness"

Either $\sigma_j^* = 0$ or $g_j(\vec{x}^*) = 0$

Lagrange multipliers for inactive constraints are 0

How do we solve for $\vec{x}^*, \vec{\lambda}^*, \vec{\sigma}^*$ that satisfy KKT?

One method (of many): Augmented Lagrange

One equality constraint:

$$\hat{f}(\vec{x}; \lambda, \mu) = \underbrace{f(\vec{x}) + \lambda h(\vec{x})}_{\text{Lagrangian}} + \underbrace{\frac{\mu}{2} h(\vec{x})^2}_{\text{External Penalty}}$$